VRPB: Key Sets, Variables, and Constraints

Sets:

L = set of linehaul customers (delivery)

B = set of backhaul customers (pickup)

$V = \{0\} \cup L \cup B$, where 0 = depot

Variables:

$x_{ij}^k = 1$ if vehicle k travels arc (i, j), else 0

$y_i^k = 1$ if customer i is served by vehicle k

Objective:

$\min \sum_{k} \sum_{(i,j)} c_{ij} x_{ij}^k$ (minimize total travel cost)

Key Constraints:

Capacity: $\sum_{i \in L} d_i y_i^k \leq Q$ (vehicle capacity not exceeded)

Precedence: linehaul visits before backhaul visits in each route

Each customer visited exactly once

Flow conservation at each node